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T11 – 05
Assignment No. 5

Aim: Tutorial on Data Exploration

1. For the given data set

57, 57, 57, 58, 63, 66, 66, 67, 67, 68, 69, 70, 70, 70, 72, 73, 75, 75, 76, 76, 78, 79, 81 find the

- a) Mean
- b) Median
- c) Mode
- d) Midrange
- e) Range
- f) All quartiles
- g) IQR
- h) Draw boxplot for the above dataset. Comment about outliers.

BI Lab Assignment 5

Q.1 a) $\text{Mean} = \frac{1590}{23} = 69.13$

b) Median $\Rightarrow$ Middle value when data is sorted in ascending order
Median = 70

c) Mode = 57, 70 (Appears 3 times)

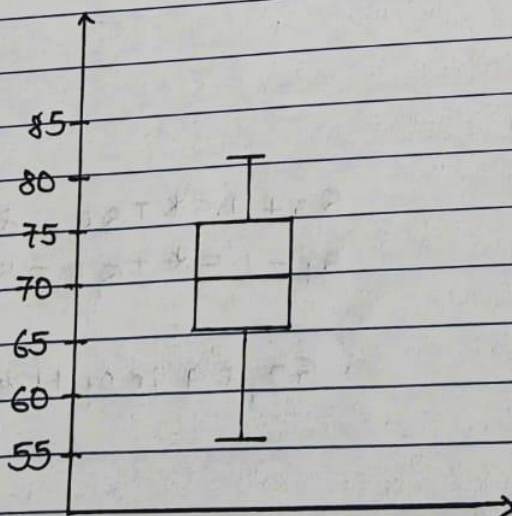
d) $\text{Midrange} = \frac{\text{Min} + \text{Max}}{2} = \frac{57 + 81}{2} = 69$

e) $\text{Range} = 81 - 57 = 24$

f) All quartiles:
 $Q_1 = 66, Q_2 = 70, Q_3 = 75$

g) $\text{IQR} = Q_3 - Q_1$
 $= 75 - 66 = 9$

h) Boxplot:



No outliers

2. 52, 57, 57, 58, 63, 66, 66, 67, 67, 68, 69, 70, 70, 70, 72, 73, 75, 75, 76, 76, 78, 79, 89 find the

a) Mean

b) Median

c) Mode

d) Midrange

e) Range

f) All quartiles

g) IQR

h) Draw boxplot for the above dataset.

Comment about outliers.

Q.2

a) Mean = $\frac{1595}{23} = 69.26$

b) Median = 70

c) Mode = 70

d) Mid range = $\frac{52 + 89}{2} = 70.5$

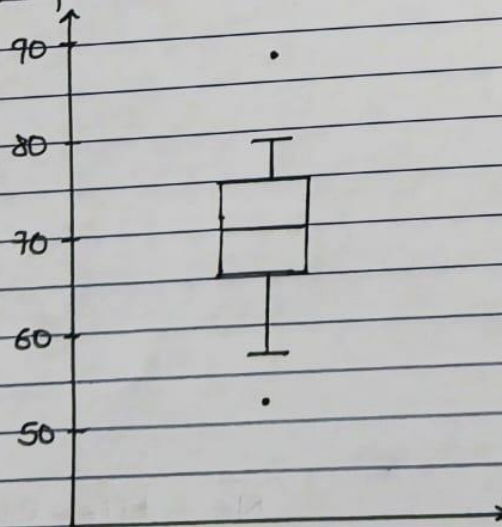
e) Range = $89 - 52 = 37$

f) All quartiles:

$Q_1 = 66, Q_2 = 70, Q_3 = 75$

g) IQR = $75 - 66 = 9$

h) Boxplot:



$Q_3 + 1.5 * IQR = 88.5$

$Q_1 - 1.5 * IQR = 52.5$

$\therefore 52, 89$ is outlier

3. Use MS-Excel to

a. Extract summary of descriptive statistics

File Home Insert Page Layout Formulas Data Review View								
H28		fx						
	A	B	C	D	E	F	G	H
1	Column1							
2								
3	Mean	29.962963						
4	Standard Error	2.49071294						
5	Median	25						
6	Mode	25						
7	Standard Deviation	12.9421241						
8	Sample Variance	167.498575						
9	Kurtosis	2.14199779						
10	Skewness	1.24575654						
11	Range	57						
12	Minimum	13						
13	Maximum	70						
14	Sum	809						
15	Count	27						
16								

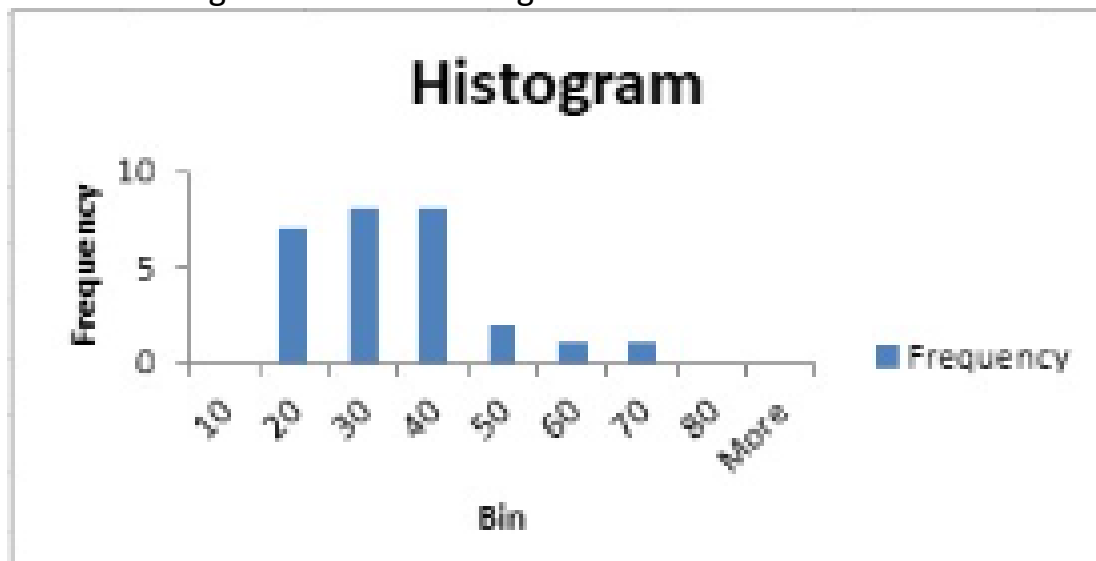
b. percentiles of 10, 20,25, 50,75, 80,90,100

File Home Insert Page Layout Formulas Data Review View								
K22		fx						
	A	B	C	D	E	F	G	H
1	Dataset	Percentile						
2	13	16						
3	15	20						
4	16	20.5						
5	16	25						
6	19	35						
7	20	35.8						
8	20	45.4						
9	21	70						
10	22							
11	22							
12	25							
13	25							
14	25							
15	25							
16	30							
17	33							
18	33							
19	35							
20	35							
21	35							
22	35							
23	36							
24	40							
25	45							
26	46							
27	52							
28	70							
29								
30								

c. Create frequency bin

N19			
	A	B	C
1	Bin	Frequency	
2	10	0	
3	20	7	
4	30	8	
5	40	8	
6	50	2	
7	60	1	
8	70	1	
9	80	0	
10	More	0	
11			
12			

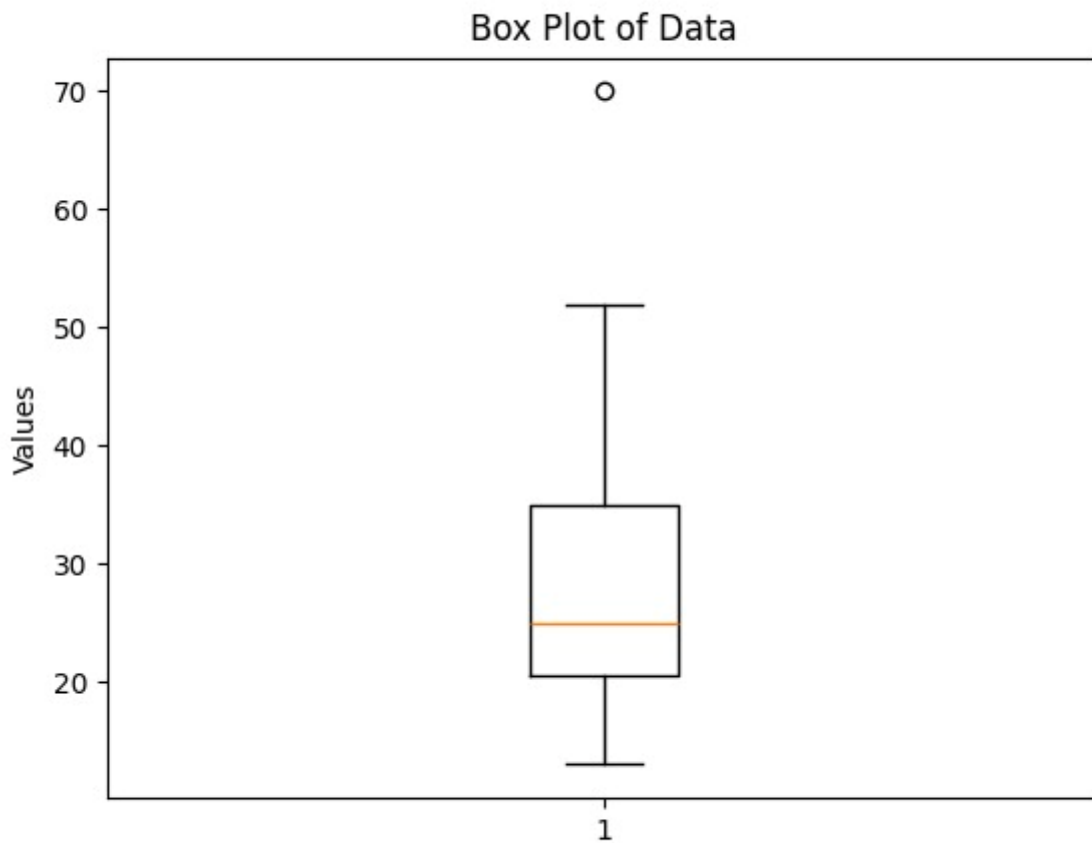
d. Draw histogram of the following dataset



e. Draw boxplot of the following dataset

13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70

Figure 1



4. Find median of grouped data

CI	F
0-10	4
10-20	6
20-30	18
30-40	34
40-50	40

Q.4

→

CI	F	CF
0-10	4	4
10-20	6	4+6=10
20-30	18	10+18=28
30-40	34	28+34=62
40-50	40	62+40=102

$$\therefore N = 102$$

$$N = 102 = \frac{51}{2}$$

Here CF just exceeding 51 is 62,

$\therefore$ Median class: 30-40

$$\therefore \text{Median} = L + \left(\frac{N/2 - CF}{F} \right) \times h$$

$$= 30 + \left(\frac{51 - 28}{34} \right) \times 10$$

$$= 36.76$$

5. Partition data into 4 bins using equi depth binning method and perform smoothing according to following methods

- a) Smoothing by bin median
- b) Smoothing by bin mean
- c) Smoothing by bin boundaries

Data 11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,30,40,45,45,45,71,72,73,75

Q.5

→ Sorting & dividing the data set equally into 4 bins:

Bin 1 = 11, 13, 13, 15, 15, 16

Bin 2 = 19, 20, 20, 20, 21, 21

Bin 3 = 22, 23, 24, 30, 40, 45

Bin 4 = 45, 45, 71, 72, 73, 75

a) Smoothing by bin median

Median of bin 1 = $(13+15)/2 = 14$

Median of bin 2 = 20, Median of bin 3 = 27

Median of bin 4 = 71.5

∴ Bin 1 = 14, 14, 14, 14, 14, 14; Bin 2 = 20, 20, 20, 20, 20, 20

Bin 3 = 27, 27, 27, 27, 27, 27; Bin 4 = 71.5, 71.5, 71.5, 71.5, 71.5, 71.5

b) Smoothing by Bin Mean

Bin 1 = Mean = 13.83, Bin 2 = Mean = 20.16

Bin 3 = Mean = 30.66, Bin 4 = Mean = 63.5

∴ Bin 1 = 13.83, 13.83, 13.83, 13.83, 13.83, 13.83

Bin 2 = 20.16, 20.16, 20.16, 20.16, 20.16, 20.16

Bin 3 = 30.66, 30.66, 30.66, 30.66, 30.66, 30.66

Bin 4 = 63.5, 63.5, 63.5, 63.5, 63.5, 63.5

c) Smoothing by bin boundaries

Extreme data points in bin are kept the same.

Data points in between are replaced by boundary values which they are close to.

Bin 1 = 11, 11, 11, 16, 16, 16

Bin 2 = 19, 19, 19, 19, 21, 21

Bin 3 = 22, 22, 22, 22, 45, 45

Bin 4 = 45, 45, 75, 75, 75, 75